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Nilus Lab Zenith: An Integrated 5-Layer Epigenomic Intelligence & Cellular Rejuvenation Discovery Engine

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ABSTRACT

Restoring youthful cellular function and rescuing multigenic disease in human tissues requires an integrated system that can decode non-coding genetic variants, model complex gene regulatory networks, predict biophysical cardiotoxicity, and optimize molecular delivery. Here we present the architecture, methodology, and computational evaluation of **Nilus Lab Zenith (v31.0)**, a computational biology platform for in silico cellular trajectory modeling. Zenith comprises a generalist scVI foundation model (1,962,128 registered human single cells, 5,858 HVGs, 4 hidden layers $\times$ 1,024 units, 64 latent dimensions; 37,234,698 trainable parameters) and a cardiac specialist scVI model (trained on **99,993 Specialist Cardiac cells from a 486,134-cell source atlas** across 14 donors, 5,009 HVGs, 2 hidden layers $\times$ 128 units, 20 latent dimensions; 3,268,462 trainable parameters), coupled with biophysical 3D structural protein prediction (Boltz-1, ESMFold) and a 512-neuron spiking electrophysiological simulation (NEUROS-X). Across five computational layers, Zenith unifies epistatic polygenic risk scoring, 201-base-pair transcription factor binding motif scanning (AlphaGenome), causal inference via Mendelian Randomisation (IVW/Egger), multi-omics factor analysis (MOFA+), structural docking (AutoDock Vina), CRISPR Prime Editor pegRNA synthesis, selective organ targeting (SORT) 5-lipid nanoparticles, reinforcement learning cocktail discovery (AlphaZen), and two-compartment PK/PD virtual clinical trials. In leave-one-donor-out cross-validation across the 54-donor uniform-chemistry PERIHEART human right atrial cohort (392,819 single nuclei), a ridge-regularized pseudobulk aging clock achieved statistically significant out-of-donor age prediction (LODO MAE = 6.97 years, Pearson $r = 0.4606$, $p = 4.57 \times 10^{-4}$; $r = 0.4936$ across all 8,560 active genes), while in silico screening of candidate reprogramming cocktails is gated by a 12-channel cardiomyocyte leaky integrate-and-fire electrophysiological simulation ($\sigma^2_{\text{ISI}} = 46.42$). Zenith bridges computational predictions to physical laboratory execution via 1-click Opentrons OT-2 robotic pipetting protocol generation and microfluidic LNP formulation.

Keywords: Cellular Rejuvenation, BiT Age Transcriptomic Clock, Epigenetic Clocks, Single-Cell Omics, Mendelian Randomisation, Prime Editing, Lipid Nanoparticles, Cardiac Electrophysiology, Autonomous AI Agents.

1.0 INTRODUCTION

The central paradigm of longevity medicine is shifting from passive disease treatment to proactive cellular rejuvenation (1, 2). While classic Yamanaka factor reprogramming (OCT4, SOX2, KLF4, c-MYC; OSKM) demonstrated that somatic cell identity is plastic (3), full reprogramming in living organisms leads to loss of functional cell identity and lethal teratoma formation (4, 5). Partial reprogramming strategies offer a promising alternative, yet they introduce significant safety bottlenecks:

- **Cardiotoxicity & Arrhythmia:** Altering ion channel expression during lineage transition can trigger lethal Long-QT syndrome, catecholaminergic polymorphic ventricular tachycardia (CPVT), or ventricular fibrillation.
- **Off-Target Delivery Sequestration:** Unmodified delivery vehicles suffer from passive hepatic trapping via ApoE protein corona opsonization ($< 10\%$ target uptake).
- **Non-Coding Causal Ambiguity:** Over 90% of GWAS SNPs reside in non-coding regulatory elements where linear models fail.

To address these challenges, we engineered **Nilus Lab Zenith** in London, UK—a closed-loop discovery instrument unifying five computational layers.

2.0 COMPUTATIONAL ARCHITECTURE

Zenith is structured as a 5-layer hierarchy processing inputs from lower modules up to autonomous optimization agents.

2.1 Layer 1: Genomic & Epigenetic Foundation

Single-Cell Latent Manifold (scVI): Embeds single-cell expression into a 20-dimensional (specialist) or 64-dimensional (foundation) latent space ($q_{\phi(z|x,s)}$). Comprises a generalist scVI foundation model trained on **1,962,128 human single cells** (37,234,698 trainable parameters) and a cardiac specialist scVI model trained on **99,993 Specialist Cardiac cells (from a 486,134-cell source atlas)** across 14 donors, Litviňuková et al. 2020; 3,268,462 trainable parameters).

Epistatic Polygenic Risk Score (PRS):

$$\begin{aligned} \text{PRS}_{\text{total}} &= \sum \beta_i G_i + \sum \sum \gamma_{ij} G_i G_j + \epsilon \\ \Delta\text{Horvath} &= 0.085 \times \text{PRS}_{\text{total}} \text{ (years)} \end{aligned}$$

AlphaGenome 201bp PWM Scanner: Evaluates 19 JASPAR 2024 pioneer TFs (GATA4, MEF2C, NKX2-5, POU5F1, SOX2, MYC). $|\Delta\text{PWM}| > 2.0$ flags disrupted binding sites.

Structural Authority & Clocks: Incorporates Boltz-1 and ESMFold 3D structure predictions, the Meyer-Schumacher BiT age transcriptomic clock (binarized scRNA regression; the Zenith implementation is uncalibrated and reports no accuracy figure), cross-modal Horvath DNA methylation projection (353 CpGs), DunedinPACE rate, and ADMET cardiotoxicity filters.

2.2 Layer 2: Causal Inference & Multi-Omics

Mendelian Randomisation (MR):

$$\hat{\beta}_{IVW} = \sum (w_i \beta_{Y,i} / \beta_{X,i}) / \sum w_i$$

Constrained to $F > 10$, MR-Egger pleiotropy intercept testing ($p > 0.05$), and Steiger directionality filtering.

MOFA+ Integration: Learns $K=10$ latent factors across scRNA, ATAC, Methylation, Proteomics, and Metabolomics. Epistatic GRN Causal Mapper builds directed DAGs (DoRothEA/GENIE3).

2.3 Layer 3: Interventions & Delivery

AutoDock Vina & Prime Editing: Computes ΔG_{bind} docking free energy and synthesizes pegRNAs ($\text{Spacer}_{20} + \text{Scaffold}_{80} + \text{PBS} + \text{RTT}$).

LNP SORT Optimizer v2: Formulates 5-component SORT lipid nanoparticles (DLin-MC3-DMA 50%, DSPC 10%, Cholesterol 38.5%, DMG-PEG2000 1.5%, DOTAP 10-20%) with a predicted $> 85\%$ cardiac capillary tropism (model output; not measured experimentally).

2.4 Layer 4: AlphaZen RL & Virtual Trials

$$\begin{aligned} R &= w_1 \Delta\text{Horvath} - w_2 \text{CardioRisk} - w_3 \text{OncogenicRisk} + w_4 \text{Yield} \\ dC_1/dt &= (D/V_1) - (k_{12} + k_{10})C_1 + k_{21}C_2 \end{aligned}$$

AlphaZen reinforcement learning PPO policy search coupled with 2-compartment PK/PD virtual trial simulations ($N=1,000$).

2.5 Layer 5: NEUROS-X Safety & Autonomous Agent

$$\begin{aligned} C_m dV_i/dt &= -g_L(V_i - E_L) + I_{\text{syn},i}(t) + I_{\text{ion},i}(t) \\ \sigma^2_{\text{ISI}} &= (1 / (N_{\text{spikes}} - 1)) \sum (t_{k+1} - t_k - \bar{t}_{\text{ISI}})^2 \end{aligned}$$

Audits electrophysiological safety on a 512 LIF spiking neuron cardiac substrate ($\sigma^2_{\text{ISI}} < 60.0$). 10-step AI agent persists runs in SQLite (``zenith_master.db``).

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ZENITH DISCOVERY BENCHMARK RESULTS (Litviňuková et al. 2020)
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Target Dataset: PERIHEART 54-Donor Replication Cohort (392,819 Nuclei) & Litviňuková Atlas (99,993 Trained / 486,134 Source
Leave-One-Donor-Out Aging Clock: LODO MAE = 6.97 Years | Pearson r = 0.4606 (p = 4.57e-4)
In Silico Screening: Exploratory 8-Factor Hypothesis Panel (Unvalidated Historical Benchmark Removed)
Electrophysiological Status (in silico): STABLE CONDUCTION – simulated ISI Variance = 46.42 (threshold < 60.0)
Predicted LNP Encapsulation Efficiency: 94.4% (model output) | >85% Predicted Cardiac Tropism
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3.0 EXPERIMENTAL RESULTS

Zenith evaluated single-cell transcriptomic correlations across the cardiac specialist atlas (99,993 trained cells from

the 486,134-cell source atlas) to nominate an exploratory 8-factor hypothesis panel (Table 1).

Table 1. Zenith Discovered 8-Factor Rejuvenation Protocol.

Priority	Gene Symbol	Functional Category	Correlation (r)	Primary Cascade & Biological Mechanism
1	SIRT1	External Sirtuin Target	Prior (N/A)*	Histone deacetylation → Chromatin remodeling → Epigenetic age reset
2	SIRT6	External Sirtuin Target	Prior (N/A)*	Histone deacetylation → Double-strand DNA repair → Telomere maintenance
3	TTN-AS1	Long Non-Coding RNA	+0.300	Titin mRNA regulation → Sarcomere Z-disc organization → Capacitance stability
4	MLIP-AS1	Long Non-Coding RNA	+0.245	Lipid metabolism modulation → Mitochondrial ATP yield → Functional contraction
5	SH3RF2	E3 Ubiquitin Ligase	+0.214	Selective ubiquitination → Fibroblast protein clearance → Proteostasis reset
6	PRKCE	Protein Kinase C ε	+0.191	Ion channel phosphorylation → Membrane stabilization → Survival signaling
7	WDFY3	Autophagy Adaptor	+0.185	Autophagosome formation → Organelle clearance → Senescence clearance
8	DDX60L	RNA Helicase	−0.184	RNA unwinding → Transcriptomic stability → Innate defense modulation

*Note: SIRT1 and SIRT6 are curated literature priors outside the 5,009-HVG count matrix. Single-cohort correlations (Litviňuková et al., n=14 donors) did not replicate in the independent 54-donor PERIHEART cohort (top-100 gene overlap r = −0.42 vCM, −0.19 atrial CM, −0.23 fibroblast); only the multi-gene pseudobulk aging clock achieved statistically significant out-of-donor validation (LODO Pearson r = 0.4606, MAE = 6.97 yr).

Safety & Robotics: NEUROS-X simulation reported ISI variance $\sigma^2_{\text{ISI}} = 46.42$ (stable conduction). Zenith exported an Opentrons OT-2 Python script (``zenith_opentrons_1785817573.py``) for automated 96-well pipetting.

4.0 DISCUSSION

Zenith v31.0 provides an exploratory in silico hypothesis-generation framework for modeling cellular reprogramming trajectories without pluripotency factors. Leveraging lncRNAs (TTN-AS1, MLIP-AS1) and sirtuins (SIRT1, SIRT6) is proposed as a candidate route to epigenetic age reversal with preserved membrane capacitance and titin structural integrity. This is an untested computational hypothesis: none of these factors has been evaluated in vitro or in vivo, and the single-cohort correlations supporting them did not replicate in the independent PERIHEART cohort (see Table 1 note).

5.0 DATA & CODE AVAILABILITY

- **Platform:** <https://www.niluslab.com/index.html>
- **Catalog:** https://www.niluslab.com/technical_catalog.html
- **Repository:** <https://github.com/alaaaldeeen1994/is-chrp-v26-zenith>

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